Healthcare/IoT Cybersecurity Testbed: Semester 2 Project Plan

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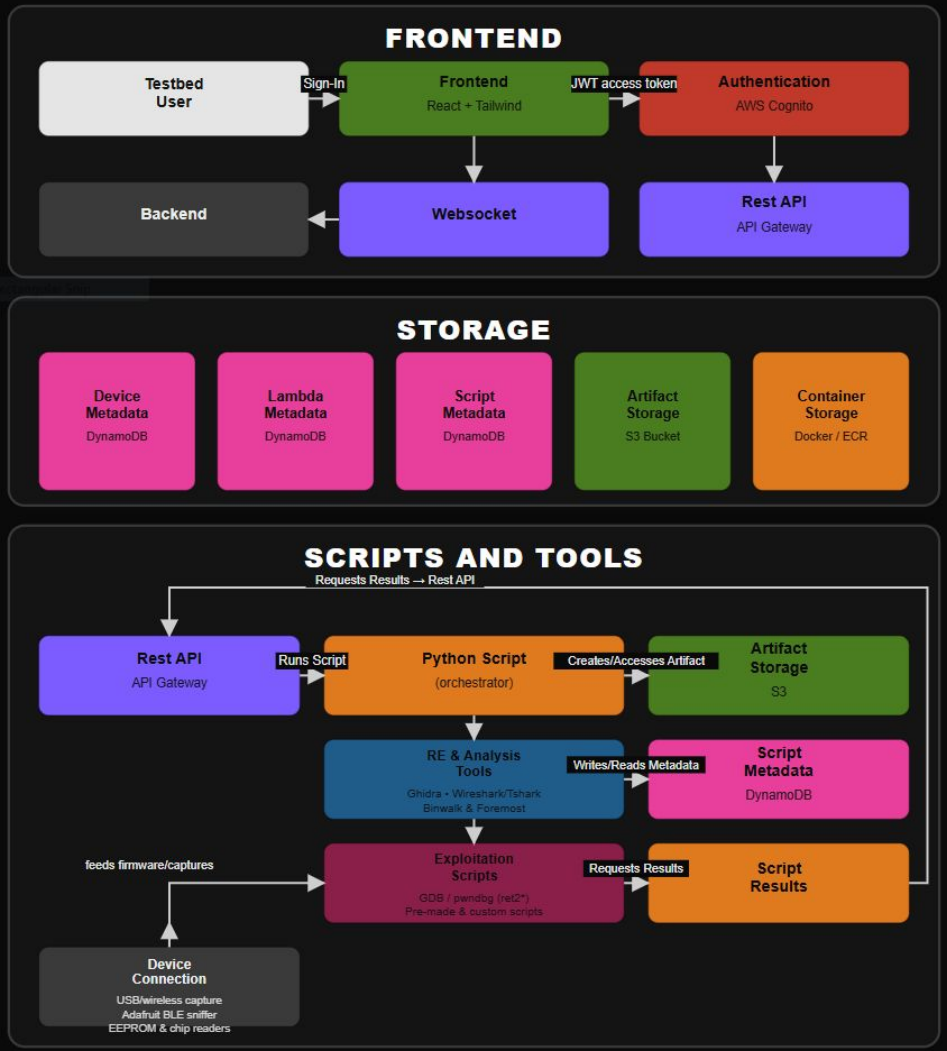
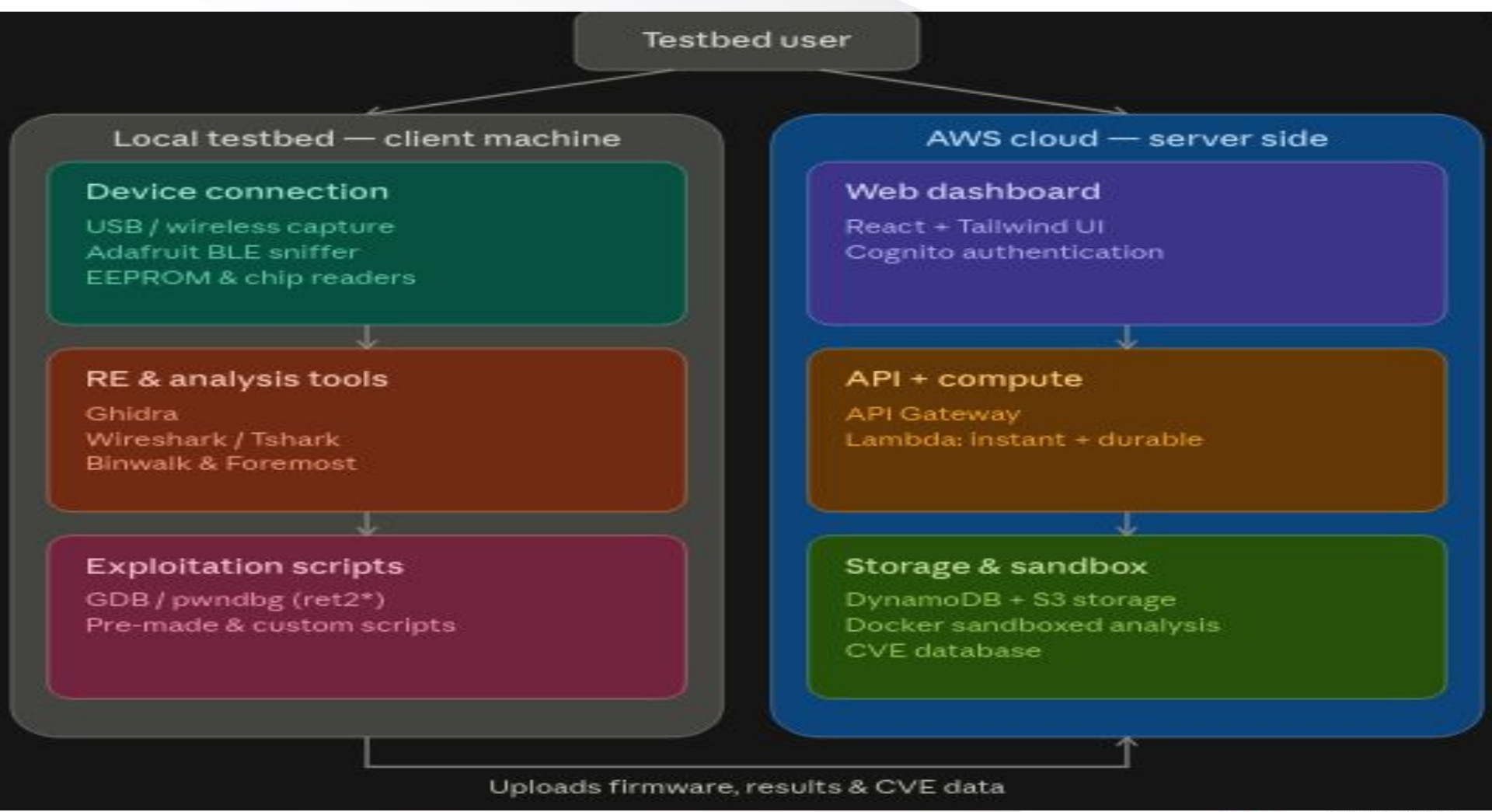
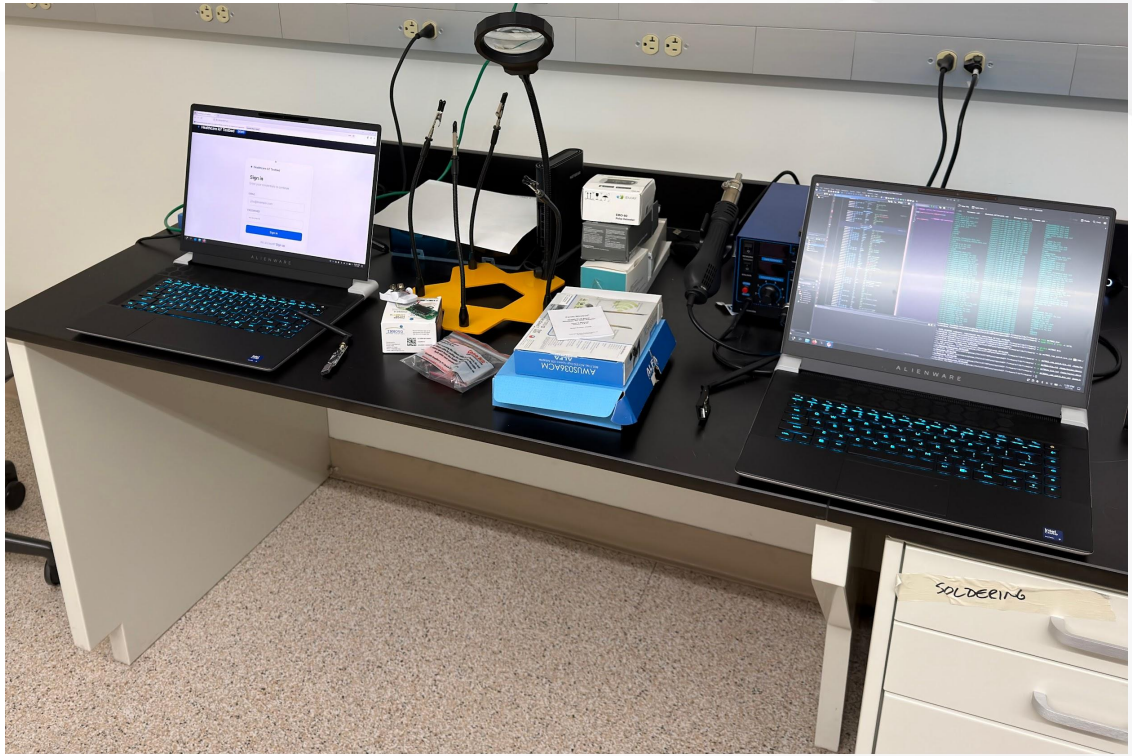

FRONTEND
Testbed User
Sign-In
Frontend
React + Tailwind
JWT access token
Authentication
AWS Cognito
Backend
Websocket
Rest API
API Gateway
STORAGE
Device Metadata
DynamoDB
Lambda Metadata
DynamoDB
Script Metadata
DynamoDB
Artifact Storage
S3 Bucket
Container Storage
Docker / ECR
SCRIPTS AND TOOLS
Requests Results → Rest API
Rest API
API Gateway
Runs Script
Python Script
(orchestrator)
Creates/Accesses Artifact
Artifact Storage
S3
RE & Analysis Tools
Ghidra + Wireshark/Tshark
Binwalk & Foremost
Writes/Reads Metadata
Script Metadata
DynamoDB
feeds firmware/captures
Exploitation Scripts
GDB / pwndbg (ret2*)
Pre-made & custom scripts
Requests Results
Script Results
Device Connection
USB/wireless capture
Adafruit BLE sniffer
EEPROM & chip readers
TOOLS
Rest API
API Gateway
Create New
New Tool Instance
Access
Container Storage
Docker sandbox
Creates
Lambda Metadata
DynamoDB
Script Metadata
DynamoDB
New Artifact
Artifact Storage
S3 Bucket
Accesses Artifact
Transmit/Receive Data
Existing Instance
Attach Existing
Durable Lambda
Instant + Durable functions
Creates New Lambda
Websocket
Tool Results
Reads Metadata
Correlates CVEs to Findings
CVE Database
Vulnerability findings correlation
Report Generation
Aggregates tool results, script results & CVE matches into a client report
SECURITY TESTING
Upload Test Data
Deploy / Run Tests
Test Executed on Device
Testbed User
Cloud Server
S3 + DynamoDB
(firmware / results / CVE)
User Device
Adafruit BLE sniffer
EEPROM & chip readers
IoT Device
Target under test
Uploads / Runs Tests on IoT Device


Testbed user
Local testbed — client machine
Device connection
USB / wireless capture
Adafruit BLE sniffer
EEPROM & chip readers
RE & analysis tools
Ghidra
Wireshark / Tshark
Binwalk & Foremost
Exploitation scripts
GDB / pwndbg (ret2*)
Pre-made & custom scripts
AWS cloud — server side
Web dashboard
React + Tailwind UI
Cognito authentication
API + compute
API Gateway
Lambda: instant + durable
Storage & sandbox
DynamoDB + S3 storage
Docker sandboxed analysis
CVE database
Uploads firmware, results & CVE data

Physical Testbed



Here is the current demo deployment of the physical testbed, featuring the virtual components, as follows:

Server-side -> Web Dashboard (Left)

Local-Side -> Analysis Tooling (Right)

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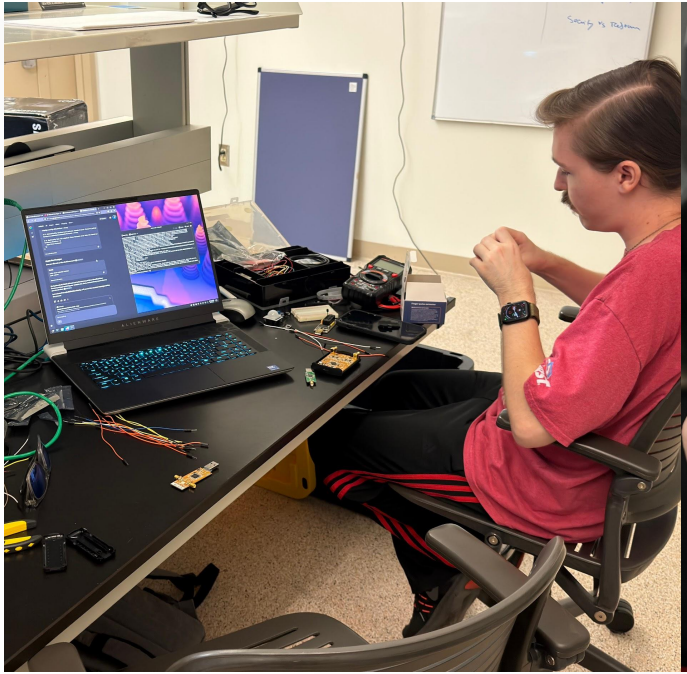
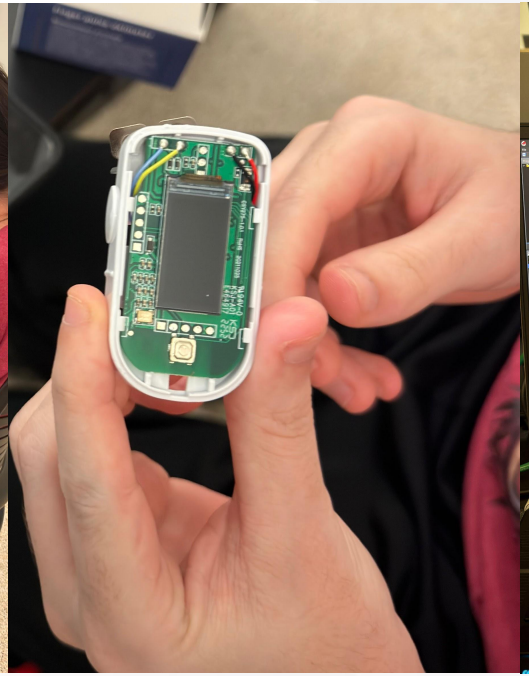
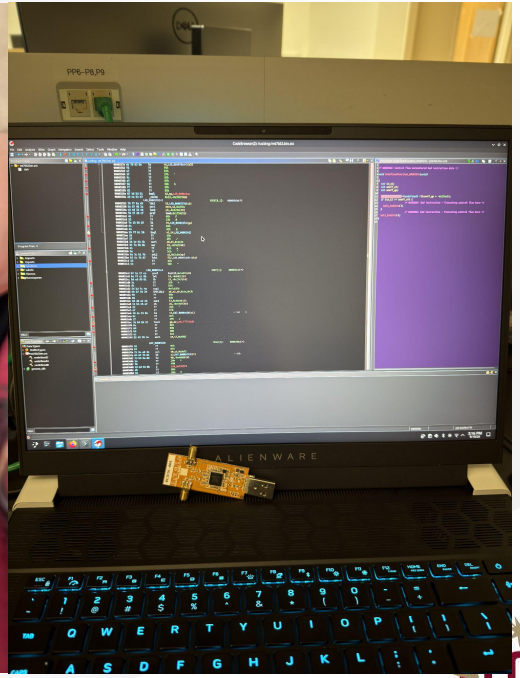
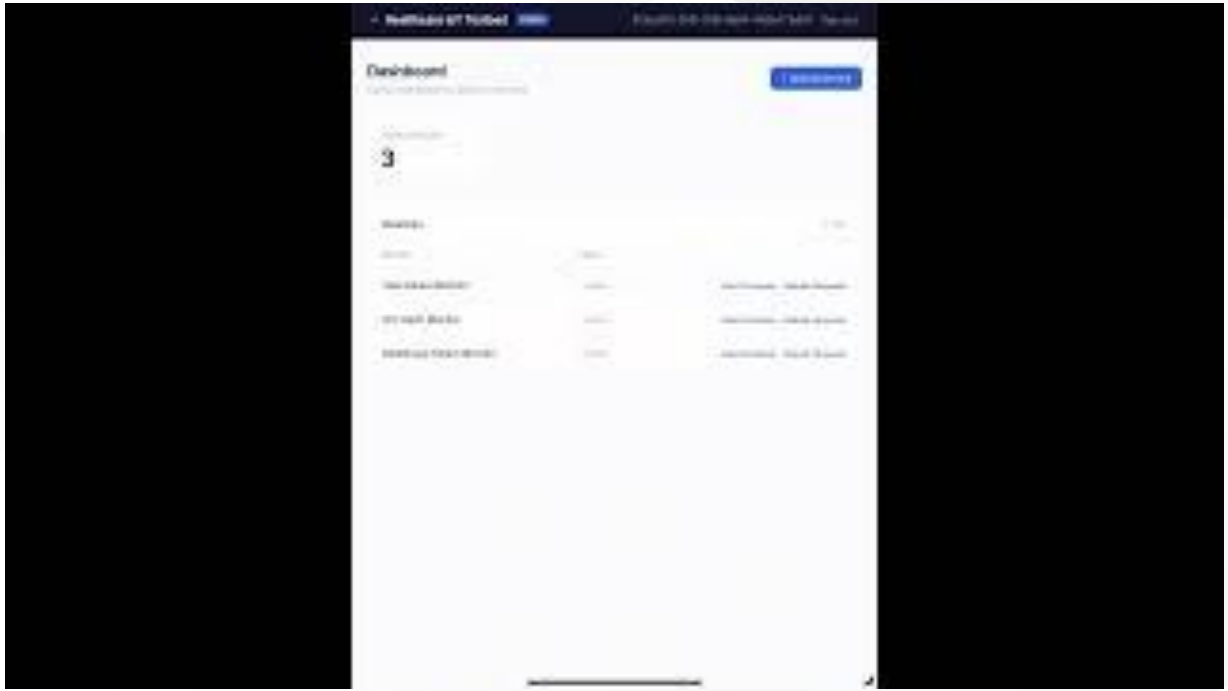
Reverse Engineering







Web Interface Demo - Device Management & Firmware Upload



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Frontend Integration- Device Management and Upload Workflow

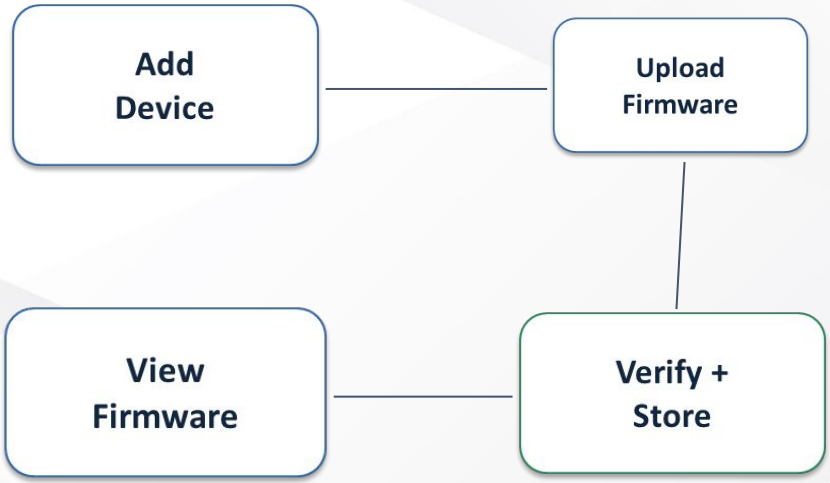
Authenticated dashboard workflow for creating and listing IoT/medical devices.

Firmware upload modal connected to the real backend and S3 upload path.

Backend confirmation step makes upload completion authoritative.

Firmware viewing/refetch workflow exposes the stored artifact state.

Retry/concurrency design protects against stale upload attempts.



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Device Creation Flow

A device is created once, then becomes the parent for firmware artifacts.

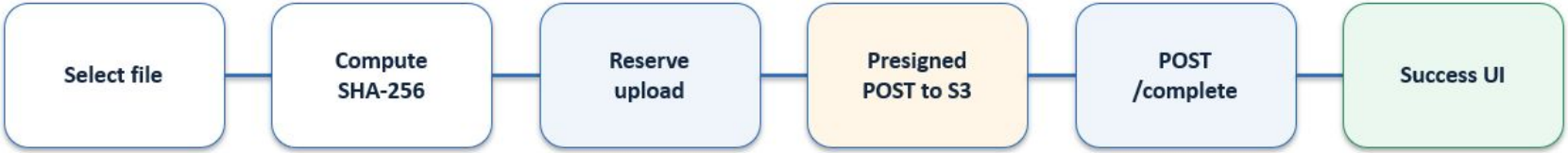
Device metadata uses DEVICE#<deviceId> / METADATA.

Relationship rows associate the authenticated user with the device.

That relationship lets later list operations return only the user's devices.

Firmware Upload: End-to-End

The key design is a two-stage upload: reserve metadata first, then upload the binary directly to S3.



25 MiB client-side size cap; Web Crypto computes SHA-256.

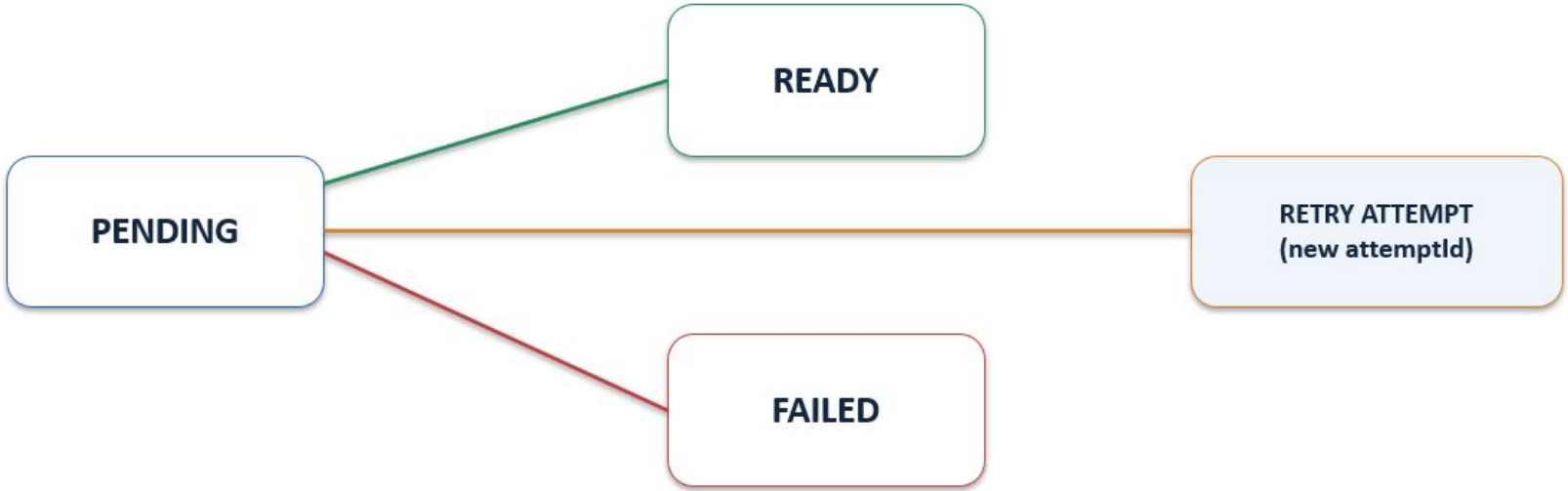
Reservation records immutable firmware metadata and creates an upload attempt.

The binary bypasses Lambda, reducing backend transfer/load.

/complete checks the uploaded S3 object against the reserved metadata.

Firmware State & Retry Semantics

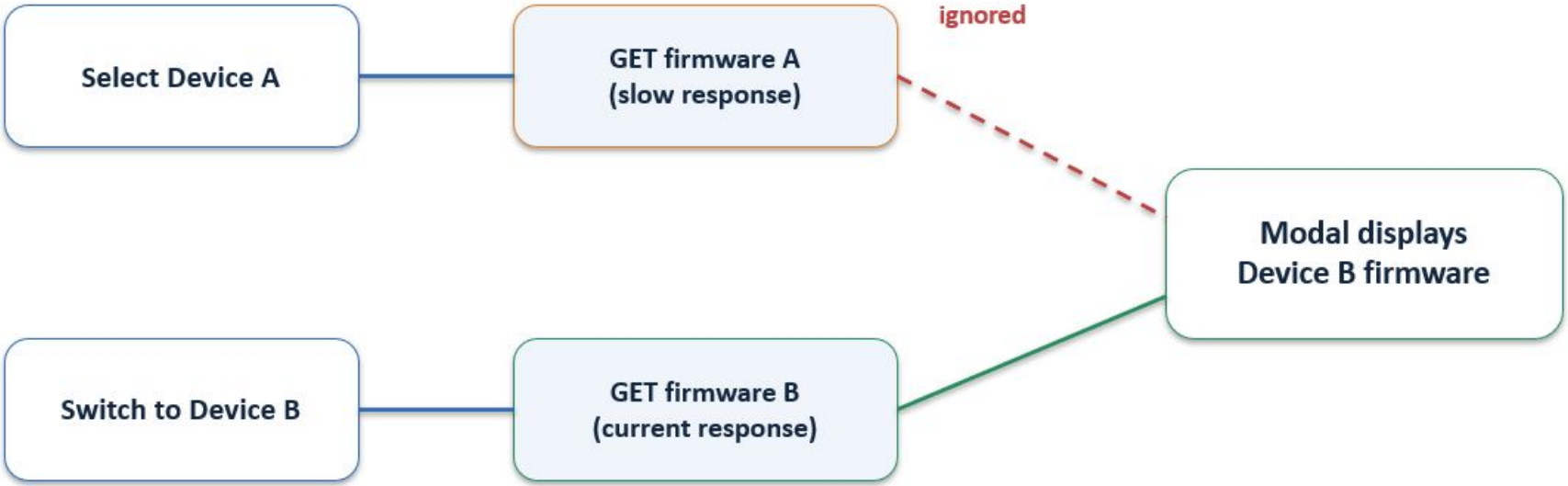
Retry is designed for an unfinished pending attempt—not to repair immutable metadata that was proven wrong.



Important: retry preserves firmwareId, filename, version, sizeBytes and sha256. retryOfAttemptId identifies which attempt the client is retrying, protecting against stale clients/concurrent attempts.

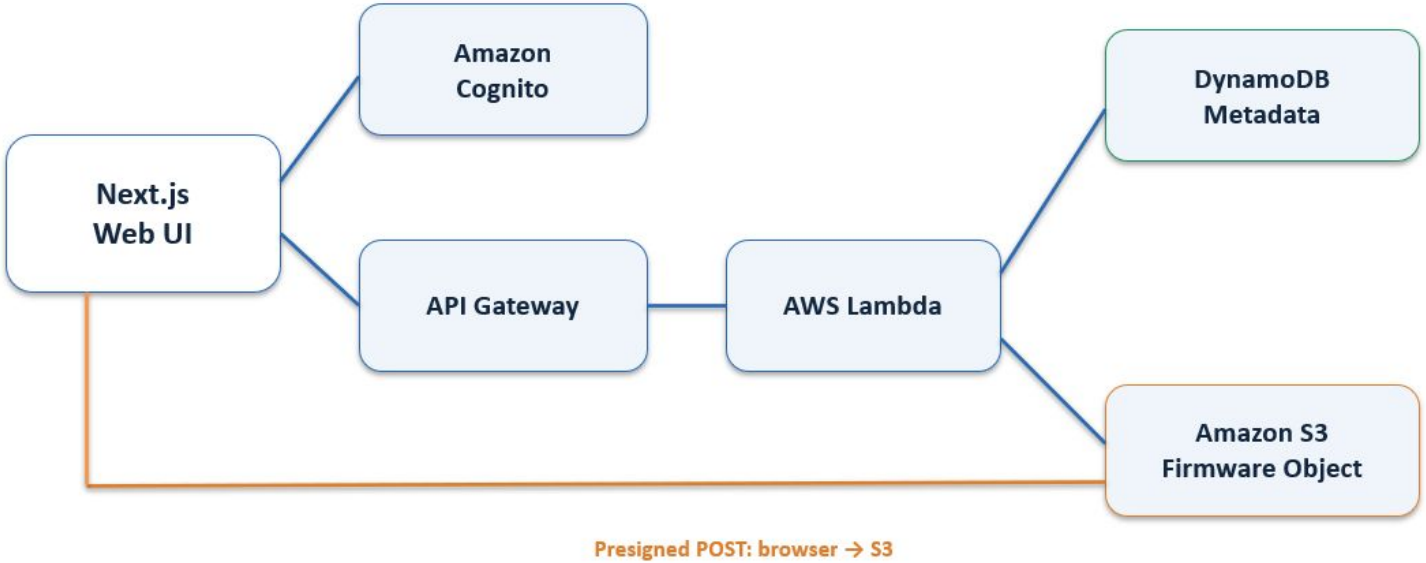
View Firmware: Avoiding Stale UI Results

The modal refetches for the selected device, and cleanup prevents an older request from overwriting a newer selection.



Frontend and AWS System Architecture

The browser does not directly own infrastructure logic; it coordinates authenticated APIs and a presigned S3 upload.



Frontend Integration - Key Features

The implementation is more than a file-upload button: the workflow preserves security, consistency, and testbed isolation.

Security: authenticated API calls; browser receives temporary presigned upload permission rather than AWS credentials.

Scalability: firmware binaries go directly to S3; Lambda manages metadata and verification.

Integrity: SHA-256 + size are reserved and later verified before the firmware becomes ready.

Concurrency: attemptId / retryOfAttemptId prevents a stale client from blindly retrying the wrong upload attempt.

Consistency: /complete is authoritative for the upload action; list/view is a fresh read for display.

Testbed relevance: each stored firmware artifact can become the reproducible input for later isolated vulnerability-analysis environments.

Key System Features

- Users can connect a variety of IoT and medical devices that use standard interfaces, wired or wireless, to the testbed, enabling full device access, interaction, and manipulation.
- Users can extract firmware directly from connected IoT and medical devices, obtaining complete firmware/binaries for detailed examination and vulnerability assessment.
- Users can analyze firmware entropy to evaluate randomness and compression characteristics
- Users can utilize industry-standard open-source tools to perform reverse-engineering tasks and comprehensive vulnerability research.
- Users can input their own CVE lists, and automatically retrieve lists of known CVEs, which are stored server-side in a database.
- Users can utilize pre-made scripts or upload custom exploitation scripts to rapidly test known, suspected, or new vulnerabilities, and potential attack vectors.
- Users can perform all connection, extraction, analysis, and exploitation tasks through an intuitive client-side graphical interface, thereby streamlining the process for both novice and experienced users.
- Users can create and upload documentation, or integrate existing documentation via GitHub, with all materials stored in server-side storage.

What makes this novel?

- A testbed that tightly integrates many disparate tools, automates firmware extraction, firmware analysis, entropy analysis, decryption, into one platform, is a novel approach to cybersecurity analysis.
- The testbed reduces the background knowledge needed for manufacturers or researchers to reverse-engineer Medical/IoT devices while unifying the reverse-engineering toolset.
- The testbed is specifically tailored for IoT and medical hardware, which is often proprietary and notoriously difficult to analyze.
- The testbed allows for direct firmware extraction from connected devices via standard interfaces and therefore streamlines the process of obtaining firmware/binaries for vulnerability assessment.
- The testbed allows for built-in entropy analysis combined with pre-existing decryption techniques to uncover encrypted or obfuscated firmware sections more efficiently.

What makes this novel?

- The testbed has support for both ready-made exploitation scripts and user-developed custom scripts provides flexibility to confirm known weaknesses or explore new attack vectors.
- The testbed will have backend/web support for supporting the local GUI that lowers the barrier to entry so that both novice and experienced users can perform complex tasks with greater ease.
- The testbed will have documentation storage and a CVE database with associated automations, accelerating the research process via targeted testing, and expedited sharing of results
- The testbed's benefits scale with the difficulty of reverse-engineering each device and contribute to a more secure future for IoT and Medical devices by making thorough examination more accessible.

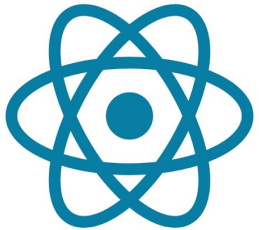
What tools we will be using?

USB EEPROM Programmer

Reverse Engineering Tools:

- Ghidra: Open source tool for binary/firmware analysis
- Wireshark/Tshark (GUI and CLI): Open source tool for network analysis
- Binwalk/Foremost CLI: Useful for examining files, extracting sub-files, and entropy analysis
- GDB/pwndbg: Useful for low-level Assembly-language reverse engineering, debugging, and vulnerability research
- Python: Ideal for creating scripts that interface with aforementioned tools via libraries and APIs
- Adafruit nRF51822-based Bluefruit LE Sniffer: USB device, drivers, and matching API for network capture (Bluetooth traffic logging as pcap files)
- Various Wireless Adapters: USB device, drivers, and matching API for network capture (Wifi/RF traffic logging as pcap files)
- EEPROM and Live Chipset Reader: Useful for pulling firmware directly off PCBS
- Soldering irons: For connecting and disconnecting electrical connections
- Solder wiring (lead or non lead): For establishing and holding electrical connections
- Copper wick: For wiping unused/excess solder
- Alligator clips: For holding PCBs and providing temporary connections where soldering is unnecessary

What tools we will be using?



Frontend and Backend Tools:

- React: Frontend of website
- TailwindCSS: Styling of website
- Terraform: Deployment of services to cloud in a reproducible fashion that allows for easy transfer of the stack to a different AWS account
- Docker: Isolates tasks for security, ease of deployment and platform management

Amazon Web Services (AWS): Provides web server hosting, database backend, and other miscellaneous compute needs listed below:

- DynamoDB: Database architecture backbone; keeps data organized, minimizes storage space complexity and computational overhead via data structuring
- S3 Object Storage: Simple artifact storage and web hosting
- Instance Lambda Functions: Used to optimize speed of repetitive tasks and minimize memory space complexity
- Durable Lambda Functions: Allows for easier control of long running cloud analysis that may involve waiting for user input
- API Gateway: Allows for a secure connection between local and cloud applications
- Cloudfront: Improved web latency and increased max traffic capacity
- Cognito: AWS managed sign on portal with authentication tokens and account management

Milestone 4 Task Matrix

Team Member	Main Responsibility Category	Sub Task 1	Sub Task 2	Sub Task 3	Sub Task 4	Sub Task 5
Justin	Reverse Engineering, Vulnerability Research, Cyber Tooling/ Demos, Local GUI, Project management	Prepare a demo of the Adafruit device and pcap file extraction using Wireshark/ Tshark	Complete CVE Research for wifi chipsets and enter said CVEs into the database for demo	Study I2C sensor and plan integration into project, such as an oximeter demo, and planning how it will interface in the main project	Respond to professors emails, communicate/ coordinate with team members, attend professor meetings, create/write most of the Project Plan and Presentation	Wipe the lab computer and install Linux (Got permission already)
Nathan	Web/Database Backend and Demos	Create the infrastructure to allow for cloud based analysis	Prepare a demo of cloud based analysis of device attack results	Integrate artifact storage and database with local and cloud infrastructure	Create secure API endpoints to allow local and cloud applications to interact	Setup router for testbed and interfacing devices in the network

Ipule	Web Front End and Cloud Backend Integration	Complete the authenticated device management workflow for adding and viewing IoT/medical devices through the web dashboard.	Integrate device creation and retrieval with AWS API Gateway, Lambda, DynamoDB, and Cognito so device records are associated with the authenticated user.	Implement the end-to-end firmware upload workflow, including firmware reservation, SHA-256 metadata, presigned S3 upload, upload completion verification, and retry support for interrupted upload attempts	Implement the firmware viewing interface so users can retrieve and display firmware metadata, upload status, filename, version, file size, and other associated information for each registered device.	Strengthen firmware upload state management and authorization by using firmware/attempt identifiers to prevent stale or conflicting retries and ensure users can access only their own devices and firmware.
Jordan	RE/VR, Cyber Tooling/ Demos, Local GUI	Continue local GUI work so users can trigger the RE/analysis/exploitation tools	Continue developing pre-made exploitation scripts for GDB/pwndbg (ret2*)	Develop pre-made exploitation scripts for Ghidra and/or integrate existing Ghidra scripts	Begin live firmware extraction using EEPROM/chipset readers	Install necessary tools on Linux machine

Milestone 5 Tasks

Ipule - Web Front End and Cloud Backend Integration

- Integrate the firmware-analysis workflow into the web dashboard so a user can select an uploaded firmware image and initiate an analysis job.
- Develop frontend interfaces for viewing analysis jobs, execution status, and generated results associated with a specific device and firmware version.
- Integrate the dashboard with the backend analysis APIs and storage architecture developed by the team, including retrieval of analysis outputs and artifacts.
- Improve device and firmware workflow state handling by displaying clear pending, ready, processing, completed, and failed states where applicable.
- Implement user interface protections and validation for analysis actions so jobs can only be started against valid firmware owned by the authenticated user.

Nathan - Web/Database Backend and Demos

- Create proof of concept docker/sidecar deployments are properly set up with the tooling of the DS18B20 I2C sensor work
- Starting working for DS18B20 I2C sensor (and similar sensor's) exploitation can be verified by a new user through their own replication of the pipeline
- Testing security of tooling and platform itself, as part of the containerization and isolation of environment

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Milestone 5 Tasks

Justin - RE/VR, Cyber Tooling/Demos, Local GUI, Project Management

- Begin integrating Adafruit and Wireshark/Tshark
- Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation
- Keep project/team members moving and ensure Professors (customers) are happy
- Maintain email/Discord communications constantly
- Create intermediate start-to-finish proof of concept demo

Jordan - RE/VR, Cyber Tooling/Demos, Local GUI

- Continue local GUI work so users can trigger the RE/analysis/exploitation tools
- Build out Ghidra-based reverse engineering workflows for extracted firmware, producing analysis outputs
- Begin integrate entropy analysis and file segmentation tools (Binwalk/Foremost)
- Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation
- Begin I2C sensor integration into project

Milestone 6 Tasks

Ipule - Web Front End and Cloud Backend Integration

- Complete the end-to-end web workflow from device registration and firmware upload through analysis execution and result retrieval.
- Integrate available reverse-engineering and vulnerability-analysis outputs into the dashboard, such as extracted firmware information, entropy results, discovered files, vulnerabilities, or generated artifacts provided by the analysis backend.
- Improve the dashboard user experience by organizing devices, firmware versions, analysis jobs, and outputs into a clear workflow suitable fo researchers and manufacturers.
- Perform integration and edge-case testing of authentication, device ownership, firmware upload/retry behavior, analysis initiation, and result retrieval.
- Prepare the web application for the final project demonstration by resolving integration issues, improving error handling, and documenting/demoing the complete user workflow.

Jordan - RE/VR, Cyber Tooling/Demos, Local GUI

- Finalize device connections, to firmware extractions, to entropy/decryption, to exploitation/analysis paths, creating multiple local pipelines that feeds results to the web backend
- Finalize local GUI work so users can trigger the RE/analysis/exploitation tools locally
- Complete and test the exploitation scripts against real device/firmware data
- Prepare demos of the cyber tooling integrations for the final presentation
    - GDB/pwndbg
    - Ghidra demo

Milestone 6 Tasks

Justin - RE/VR, Cyber Tooling/Demos, Local GUI, Project Management

- Finalize CVE entries and RE documentation for the final demo
- Prepare demos of the cyber tooling integrations for the final presentation
    - Adafruit/Wireshark
    - Binwalk/Foremost
- Prepare a full end-to-end RE demo (through aforementioned local pipelines) for the final presentation as a video recording
- Keep project/team members moving to ensure Professors are satisfied with progress
- Maintain email/Discord communications constantly with team members and Professors

Nathan

- Finalize AWS permissions and backend security using least-privilege access.
- Complete integration between local analysis tools, cloud services, S3, DynamoDB, and the web application.
- Finalize and test containerized cloud analysis and isolation.
- Perform end-to-end testing and prepare the cloud/backend workflow for the final demo.

Thanks for listening!
Questions?

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